

SIX MONTH INDUSTRIAL TRAINING CSD 481 MID TERM REPORT ON

"FEASTFLOW"

Submitted in partial fulfillment of the requirement for award of Degree
of

**BACHELOR OF TECHNOLOGY
(COMPUTER SCIENCE AND ENGINEERING)**



SESSION: 2022–26

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DECLARATION

I, "**Aeronbir Singh Brar**", declare that the work presented in this project titled "***FeastFlow***", submitted to the Department of Engineering & Technology, Guru Nanak Dev University Regional Campus, Jalandhar for the award of degree of Bachelor of Technology in Computer Science and Engineering, is my original work. I have not plagiarized or submitted the same work for the award of any other degree. In case this undertaking is found incorrect, I accept that my degree may be unconditionally withdrawn.

May, 2026

Aeronbir Singh Brar

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CERTIFICATE

Certified that the work contained in this report of project titled “***FeastFlow***”, by “**Aeronbir Singh Brar**”, has been carried out under my supervision.

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ACKNOWLEDGEMENT

I, "Aeronbir Singh Brar", am highly grateful to **Dr. Deep Kamal Kaur Randhawa, Professor & Head, Department of Engineering and Technology, Guru Nanak Dev University Regional Campus, Jalandhar**, for providing this opportunity to carry out the present report/work. The consistent guidance and encouragement received from **Dr. Sheetal Kalra, our project Faculty Mentor**, has been of great help in carrying out the present work and is acknowledged with reverential thanks. Without her wise counsel and able guidance, it would have been difficult to complete the report in this manner.

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CERTIFICATE FROM THE ORGANIZATION

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THIS CERTIFICATE IS PROUDLY PRESENTED TO
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FOR SUCCESSFULLY COMPLETING TRAINING COURSE IN
FULL STACK DEVELOPMENT

FROM 21 DEC 2025 TO 21 MAY 2026







Instructor's Signature


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CHAPTER 1

INTRODUCTION TO PROJECT

1.1 Introduction & Scope of Project

In recent years, the rapid growth of internet technologies and digital platforms has significantly transformed the way businesses operate across various industries. One of the sectors that has experienced a major shift is the food and restaurant industry. Traditionally, restaurants relied on manual processes such as dine-in services, handwritten orders, and telephonic bookings. While these methods served their purpose in the past, they are no longer sufficient to meet the demands of modern customers.

Today's consumers expect convenience, speed, and accessibility. With the widespread use of smartphones and the internet, customers prefer ordering food online rather than visiting restaurants physically or placing phone calls. This shift has led to the rise of digital food ordering systems and e-commerce platforms tailored specifically for restaurants.

The Restaurant E-Commerce Website with Intelligent Order Management is designed as a full-stack web application that aims to digitalize and automate the restaurant ordering process. The system provides a centralized online platform where customers can explore menus, select food items, manage their cart, and place orders efficiently. At the same time, it provides restaurant administrators with tools to manage menu items, track orders, and maintain customer data.

Need for Digital Transformation in Restaurants:

The traditional restaurant management system suffers from several limitations:

- High dependency on manual labor
- Increased chances of order miscommunication
- Long waiting times during peak hours
- Lack of proper record-keeping
- Limited accessibility for customers

To overcome these challenges, a digital system is necessary. The proposed project addresses these issues by introducing automation, accuracy, and efficiency into restaurant operations.

Scope of the Project:

The scope of this project covers the design, development, and deployment of a web-based restaurant ordering system. It includes the following major components:

1. User Interface Development

- Development of a responsive and user-friendly interface
- Ensuring compatibility across devices (desktop, tablet, mobile)
- Designing intuitive navigation for better user experience

2. Backend System Development

- Creation of RESTful APIs for handling business logic
- Implementation of server-side operations using Node.js and Express.js
- Handling user authentication, order processing, and data management

3. Database Management

- Use of MongoDB (NoSQL database) for storing application data
- Efficient storage of:
 - User information
 - Product details
 - Orders
 - Cart data

4. Security Implementation

- Secure login system
- Password encryption
- Role-based access control (admin vs user)

5. Testing and Deployment

- Testing system functionality
- Debugging errors
- Deploying application to a live environment

Limitations of the Project:

While the system provides essential features, it has certain limitations:

- Does not include real-time delivery tracking

- Payment gateway integration is not implemented (can be added later)
- Limited to single restaurant system
- Does not include AI-based recommendations (future enhancement)

Future Scope of the Project:

The project has strong potential for expansion and improvement. Future enhancements may include:

- Integration of online payment systems (Paytm, Google Pay)
- Implementation of real-time order tracking
- Development of mobile applications (Android/iOS)
- Addition of AI-based food recommendation systems
- Expansion to a multi-restaurant platform

1.2 Description of Project

The Restaurant E-Commerce Website is a web-based platform that facilitates interaction between customers and restaurant administrators through an online interface. The system is designed using modern web technologies and follows a client-server architecture.

Overview of System Functionality:

The system operates through two main modules:

1. Customer Module

The customer module provides an interactive and easy-to-use interface for users.

Key Features:

- User Registration and Login
Users can create accounts and log in securely to access personalized features.
- Menu Browsing
Users can view food items categorized into different sections such as snacks, beverages, main course, etc.
- Product Details View
Each item includes:
 - Name
 - Price
 - Description
 - Image

- **Cart Management**
Users can:
 - Add items
 - Remove items
 - Update quantities
- **Order Placement**
Users can place orders through a simple checkout process.
- **Order History**
Users can track their previous orders.

2. Admin Module

The admin module provides full control over the system.

Key Features:

- **Menu Management**
 - Add new food items
 - Update existing items
 - Delete items
- **Category Management**
 - Organize products into categories
- **Order Management**
 - View all orders
 - Update order status
- **System Monitoring**
 - Track system activity

System Architecture:

The system follows a three-tier architecture, which ensures separation of concerns:

1. Presentation Layer (Frontend)

- Built using React.js
- Handles user interface and interactions

2. Application Layer (Backend)

- Built using Node.js and Express.js
- Handles business logic and API requests

3. Data Layer (Database)

- MongoDB used for storing data
- Provides fast and flexible data retrieval

Working of the System:

The working of the system can be summarized as follows:

1. User logs into the system
2. User browses menu items
3. User adds items to cart
4. User places order
5. Order is stored in database
6. Admin views and processes order

1.3 Objectives of the Project

The objectives of this project are defined using the SMART framework to ensure clarity and effectiveness.

1.3.1 Specific Objectives

- To develop a complete restaurant e-commerce system
- To automate manual ordering processes
- To create a user-friendly interface
- To implement secure login and authentication

1.3.2 Measurable Objectives

- Reduce order processing errors by at least 70%
- Improve customer satisfaction through faster service
- Ensure system uptime and reliability

1.3.3 Achievable Objectives

- Use MERN stack technologies for efficient development
- Utilize existing frameworks and tools
- Implement features within available resources

1.3.4 Relevant Objectives

- Aligns with current industry trends
- Solves real-world restaurant problems

- Enhances digital transformation

1.3.5 Time-Bound Objectives

The project is planned to be completed within six months:

- Month 1: Requirement analysis and planning
- Month 2: System design
- Month 3–4: Development
- Month 5: Testing
- Month 6: Deployment

Learning Outcomes:

Through this project, the following skills are developed:

- Full-stack web development
- API design and integration
- Database management
- Debugging and testing
- Version control using Git

CHAPTER 2

REQUIREMENT ANALYSIS AND LITERATURE SURVEY

2.1 Existing Projects of Similar Characteristics

Before designing the proposed system, a detailed study of existing systems and similar projects was conducted. This analysis helps in understanding current solutions, identifying their limitations, and designing a better system.

2.1.1 Traditional Restaurant Systems

Traditional restaurant systems rely heavily on manual processes such as:

- Taking orders verbally or on paper
- Manual billing
- Direct communication between customer and staff

Limitations of Traditional Systems:

Despite being widely used, these systems have several drawbacks:

- High Error Rate:
Orders can be misheard or incorrectly written
- Time-Consuming:
Manual processes slow down service, especially during peak hours
- No Data Storage:
No record of customer preferences or order history
- Limited Accessibility:
Customers must physically visit or call the restaurant
- Inefficient Management:
Difficult to track orders and inventory

2.1.2 Third-Party Food Delivery Platforms

Platforms such as Swiggy, Zomato, and Uber Eats have revolutionized food ordering.

Advantages

- Easy online ordering
- Wide customer reach
- Real-time order tracking
- Digital payment support

Disadvantages

- High Commission Charges (20–30%)
- Reduced profit margins for restaurants
- Dependency on third-party services
- Limited control over branding
- Restricted customer data access

2.1.3 Existing E-Commerce Website Templates

Many developers use generic e-commerce templates for building food ordering systems.

Limitations:

- Lack of customization for restaurant workflows
- No optimized order processing logic
- Limited scalability
- Basic UI/UX

Conclusion of Existing Systems Study:

The analysis of existing systems clearly shows that:

- Traditional systems lack efficiency
- Third-party platforms reduce business control
- Generic solutions lack specialization

Therefore, there is a need for a custom-built, independent restaurant system.

2.2 Need to Develop this Project

The development of the Restaurant E-Commerce Website is justified by multiple factors.

2.2.1 Operational Need

- Automate manual processes
- Reduce human errors
- Improve order accuracy
- Speed up service

2.2.2 Business Need

- Eliminate dependency on third-party platforms
- Avoid commission charges
- Improve profit margins
- Maintain direct customer relationship

2.2.3 Customer Need

- Convenience of ordering from anywhere
- Faster service
- Better user experience
- Access to order history

2.2.4 Technological Need

- Adoption of modern technologies
- Centralized data management
- Scalable system architecture

2.2.5 Competitive Need

In a competitive market:

- Restaurants need digital presence
- Customers prefer online ordering
- Faster systems attract more users

Conclusion:

The proposed system is necessary to:

- Improve operational efficiency
- Enhance customer satisfaction
- Increase business profitability

2.3 Course of Action

The development of the system follows a structured and systematic approach.

Step 1: Requirement Analysis

- Identify system users (Customer, Admin)
- Define system functionalities
- Understand constraints

Step 2: System Design

- Design system architecture
- Define modules and components
- Plan data flow

Step 3: Database Design

- Identify entities:
 - Users
 - Products
 - Orders
- Define relationships
- Optimize schema

Step 4: Backend Development

- Setup Node.js server
- Develop REST APIs
- Implement authentication

Step 5: Frontend Development

- Design UI using React
- Create reusable components
- Integrate APIs

Step 6: Integration

- Connect frontend and backend
- Handle data exchange
- Manage errors

Step 7: Testing

- Functional testing
- API testing
- UI testing

Step 8: Deployment

- Prepare production build
- Deploy application
- Monitor performance

Development Methodology:

The project follows an iterative development approach, where:

- Features are developed in phases
- Continuous testing is performed
- Improvements are made incrementally

2.4 Literature Survey

JOURNAL:

[1] A. Kumar and S. Gupta, "Secure Role-Based Access Control in Web Applications," International Journal of Computer Applications, vol. 183, no. 12, pp. 25–31, 2021.

This paper discusses the implementation of Role-Based Access Control (RBAC) mechanisms in modern web applications. The authors explain how authentication and authorization layers improve system security and prevent unauthorized data access. They highlight the use of JWT tokens and middleware-based validation for maintaining secure sessions. The study also emphasizes password hashing and encrypted communication protocols.

This research helped in understanding secure authentication systems and implementing role-based access control within the Restaurant E-Commerce Website.

ONLINE JOURNAL:

[2] R. Patel and M. Shah. (2022). Real-Time Web Applications Using WebSocket Technology. IEEE Access. [Online]. Available: <https://ieeexplore.ieee.org>

This online journal article explores real-time communication frameworks using WebSocket and bidirectional data transfer protocols. The authors explain how real-time updates significantly improve user interaction in e-commerce and food ordering systems. The research also discusses scalability challenges and concurrent user handling.

The findings of this study influenced the implementation of real-time order tracking and live updates within the Restaurant E-Commerce Website using modern frontend-backend communication techniques.

BOOK:

[3] U.K. Roy, Web Technologies.

This book provides foundational knowledge of HTML, CSS, JavaScript, and modern web technologies. It explains client-server communication, webpage structuring, responsive design, and scripting techniques. The concepts explained in this book helped in designing the frontend structure and responsive user interface of the Restaurant E-Commerce Website.

[4] D. Herron, Node.js Web Development.

This book explains backend development using Node.js and covers server-side programming, asynchronous operations, routing, and REST API development. It also discusses middleware integration and scalable backend architecture. The concepts discussed in this book helped in developing the backend server and APIs for the Restaurant E-Commerce Website.

[5] A. Banks and E. Porcello, Learning React.

This book explains component-based frontend development using React.js. It discusses Virtual DOM, state management, hooks, routing, and reusable UI components. The architectural principles described in this book helped in developing dynamic and responsive frontend modules for the Restaurant E-Commerce Website.

[6] K. Chodorow, MongoDB: The Definitive Guide.

This book provides detailed insights into MongoDB and NoSQL database management systems. It explains JSON-based document storage, schema flexibility, indexing, and scalability. The concepts discussed in this book helped in designing efficient database collections and relationships for the Restaurant E-Commerce Website.

[7] I. Sommerville, Software Engineering.

This book explains software engineering methodologies, SDLC models, system planning, testing, and project management concepts. The techniques discussed in this book helped in understanding project planning, requirement analysis, and system design methodologies used during the development of the Restaurant E-Commerce Website.

CONFERENCE PAPER:

[8] S. Mehta and P. Jain, "Scalable MERN Stack Applications for E-Commerce Platforms," in Proceedings of the International Conference on Emerging Technologies in Computing, 2023, pp. 112–118.

This conference paper presents the use of the MERN stack (MongoDB, Express.js, React.js, Node.js) in building scalable and high-performance web applications. The authors explain how single-page applications enhance user experience and how REST APIs improve modularity and integration capabilities.

The concepts discussed in this paper helped justify the use of the MERN stack for developing the Restaurant E-Commerce Website as a responsive and scalable food ordering platform.

WEBSITE REFERENCE:

[9] React.js Official Documentation. [Online]. Available: <https://react.dev>

The official React.js documentation provides detailed guidance on component creation, hooks, state management, routing, and frontend optimization. This resource was used extensively while designing dynamic user interfaces and reusable components for the Restaurant E-Commerce Website.

[10] Node.js Official Documentation. [Online]. Available: <https://nodejs.org>

The Node.js documentation explains server-side JavaScript execution, asynchronous programming, event-driven architecture, and backend runtime management. This resource helped in understanding backend development concepts and API handling.

[11] MongoDB Documentation. [Online]. Available: <https://www.mongodb.com/docs>

The MongoDB documentation provides guidance on database schema design, CRUD operations, indexing, aggregation pipelines, and cloud database management. This resource was used while implementing the database layer of the Restaurant E-Commerce Website.

[12] MDN Web Docs (Mozilla Developer Network). [Online]. Available: <https://developer.mozilla.org>

MDN Web Docs provide extensive learning material on HTML, CSS, JavaScript, and modern web APIs. These resources were referred to for frontend development concepts, responsive design practices, and debugging techniques used during project development.

CHAPTER 3

PLANNING OF THE PROJECT

3.1 Objectives of the Project

The planning phase of the Restaurant E-Commerce Website focuses on defining clear, structured, and achievable objectives to ensure successful system development. These objectives guide the design, implementation, and deployment of the project.

3.1.1 Functional Objectives

The functional objectives define the expected capabilities of the system:

- To develop a complete online food ordering system
- To enable users to register, login, and manage their accounts securely
- To allow users to browse menu items categorized efficiently
- To implement a dynamic cart system for adding and removing items
- To provide a smooth and efficient order placement process
- To develop an admin panel for managing products and orders

3.1.2 Technical Objectives

The technical objectives ensure that the system is built using modern and scalable technologies:

- To design a three-tier architecture (Frontend, Backend, Database)
- To implement RESTful APIs for communication between client and server
- To develop a responsive frontend using React.js
- To use Node.js and Express.js for backend development
- To design a NoSQL database using MongoDB
- To ensure modular and maintainable code structure

3.1.3 Performance Objectives

Performance objectives ensure the system runs efficiently:

- Reduce API response time to less than 2 seconds
- Optimize frontend rendering for smooth user experience

- Handle multiple users simultaneously without lag
- Minimize unnecessary API calls

3.1.4 Security Objectives

Security is critical for protecting user data and system integrity:

- Implement secure password hashing using bcrypt
- Use JWT (JSON Web Tokens) for authentication
- Restrict admin access using role-based authorization
- Prevent unauthorized API access

3.1.5 Usability Objectives

- Provide a simple and intuitive user interface
- Ensure easy navigation across pages
- Maintain consistency in UI design
- Improve overall user experience

3.2 Planned Steps to Achieve Above Objectives

The project follows a structured approach inspired by the Software Development Life Cycle (SDLC).

3.2.1 Development Model Used

The project uses an Iterative Development Model, where:

- The system is developed in phases
- Each phase is tested before moving forward
- Improvements are made continuously

Advantages of Iterative Model:

- Early detection of errors
- Continuous improvement
- Flexible to changes
- Faster development cycles

3.3 System Architecture Planning

The system is designed using a three-tier architecture, which ensures separation of concerns and scalability.

1. Presentation Layer (Frontend)

- Built using React.js
- Handles user interaction
- Displays data to users

2. Application Layer (Backend)

- Built using Node.js and Express.js
- Processes business logic
- Handles API requests

3. Data Layer (Database)

- MongoDB is used
- Stores all application data
- Ensures fast data retrieval

Advantages of This Architecture:

- Scalability
- Maintainability
- Clear separation of responsibilities

3.2.2 Detailed Stepwise Planning

Step 1: Requirement Analysis

- Identify system users (Customer, Admin)
- Define use cases:
 - Login/Register
 - Browse menu
 - Place order
- Identify system constraints

Step 2: System Design

- Define architecture
- Plan data flow
- Identify modules

Step 3: Database Design

- Create collections:
 - Users
 - Products
 - Orders
 - Cart
- Define relationships between entities

Step 4: API Design

Design RESTful APIs:

Method	Endpoint	Purpose
POST	/register	Register user
POST	/login	Login user
GET	/products	Fetch menu
POST	/cart	Add item
POST	/order	Place order

Step 5: Frontend Development

- Build UI components:
 - Navbar
 - Product list
 - Cart page
- Integrate APIs using Axios
- Ensure responsive design

Step 6: Backend Development

- Setup Express server
- Implement routes and controllers

- Connect MongoDB
- Implement middleware

Step 7: Integration Phase

- Connect frontend with backend
- Test API responses
- Handle errors

Step 8: Testing

- Test APIs using Postman
- Perform UI testing
- Debug errors

Step 9: Deployment

- Build production version
- Deploy frontend and backend
- Monitor system

3.2.3 Risk Analysis and Mitigation

Every project involves risks that must be managed.

Risk	Description	Mitigation
Data Loss	Loss of user/order data	Database backups
Security Breach	Unauthorized access	Encryption, authentication
Server Failure	Backend crash	Error handling, monitoring
Performance Issues	Slow response	Optimization
Bugs	System errors	Testing

3.2.4 Project Timeline (6-Month Plan)

Month 1: Planning

- Requirement analysis
- Literature survey

Month 2: Design

- System architecture

- Database design

Month 3: Backend Development

- API development
- Authentication system

Month 4: Frontend Development

- UI design
- API integration

Month 5: Testing

- Debugging
- Performance optimization

Month 6: Deployment

- Final deployment
- Documentation

3.2.5 Feasibility Analysis

Technical Feasibility

- MERN stack is suitable
- Tools are available

Economic Feasibility

- Low-cost development
- Open-source tools

Operational Feasibility

- Easy to use system
- Minimal training required

CHAPTER 4

HARDWARE AND SOFTWARE REQUIREMENTS FOR THE PROJECT

4.1 Hardware Requirements

The hardware requirements define the physical components needed for the development, testing, and deployment of the Restaurant E-Commerce Website. These requirements ensure smooth execution of the application and efficient development workflow.

4.1.1 Development System

A computer system is required for coding, testing, and running the application.

Minimum Requirements:

- Laptop/Desktop
- RAM: 4 GB
- Processor: Intel i3 or equivalent
- Storage: 10 GB free space

Recommended Requirements:

- RAM: 8 GB or higher
- Processor: Intel i5 or higher
- SSD storage for faster performance

4.1.2 Processor Requirements

The processor plays an important role in executing multiple processes simultaneously.

- Minimum: Intel i3
- Recommended: Intel i5/i7

Importance:

- Faster code compilation
- Better handling of backend processes
- Smooth multitasking during development

4.1.3 Memory (RAM)

RAM is required for running development tools, browser, and server simultaneously.

- Minimum: 4 GB
- Recommended: 8 GB

Usage of RAM:

- Running Node.js server
- Running React development server
- Running MongoDB database
- Browser testing

4.1.4 Storage Requirements

- Minimum: 10 GB free storage
- Recommended: 20 GB or more

Used for:

- Source code
- Node modules (large size)
- Database storage
- Logs and temporary files

4.1.5 Network Requirements

- Stable internet connection
- Required for:
 - Installing dependencies (npm)
 - API testing
 - Deployment
 - Version control (GitHub)

4.1.6 End-User Requirements

For users accessing the application:

- Smartphone / Laptop
- Internet connection
- Web browser

4.2 Software Requirements

The software requirements define the technologies used for building the system.

4.2.1 Frontend Technologies

HTML5

- Provides structure of web pages
- Defines elements like forms, buttons, inputs

CSS3

- Used for styling and layout
- Enables responsive design

JavaScript (ES6+)

- Adds interactivity
- Handles client-side logic

React.js

- JavaScript library for UI development
- Uses component-based architecture

Advantages:

- Reusable components
- Fast rendering using Virtual DOM
- Easy state management

4.2.2 Backend Technologies

Node.js

- JavaScript runtime environment
- Executes server-side code

Features:

- Event-driven architecture
- Non-blocking I/O
- High performance

Express.js

- Web framework for Node.js
- Simplifies API development

Features:

- Routing
- Middleware support
- Easy integration

4.2.3 Database Technology**MongoDB**

MongoDB is a NoSQL database used for storing application data.

Features:

- Stores data in JSON-like format
- Flexible schema
- High scalability

Advantages:

- Easy integration with Node.js
- Faster data retrieval
- Suitable for dynamic applications

4.3 Tools Used for Development

This section explains the tools used during development.

4.3.1 Visual Studio Code (VS Code)

- Lightweight code editor
- Supports multiple programming languages
- Provides extensions for development

4.3.2 Git and GitHub

- Git: Version control system
- GitHub: Code hosting platform

Benefits:

- Track code changes
- Collaborate with team
- Backup project

4.3.3 Postman

- Used for testing APIs
- Helps send HTTP requests

Features:

- Test GET, POST, PUT, DELETE
- Debug API responses

4.3.4 Web Browser (Chrome/Firefox)

- Used for running application
- Developer tools for debugging

4.3.5 MongoDB Compass

- GUI tool for MongoDB
- Helps visualize database

4.3.6 Package Manager (npm)

- Manages project dependencies
- Installs libraries and tools

4.4 Techniques Used for Development

This section is very important as it shows the technical methods used in the project.

4.4.1 RESTful API Architecture

The system uses REST (Representational State Transfer) architecture.

Key Principles:

- Stateless communication
- Client-server separation
- Resource-based URLs

HTTP Methods Used

- GET → Retrieve data
- POST → Create data
- PUT → Update data
- DELETE → Remove data

4.4.2 CRUD Operations

CRUD operations are the foundation of the application:

- Create → Add new users/orders
- Read → Fetch products/menu
- Update → Modify cart
- Delete → Remove items

4.4.3 Authentication and Authorization

Authentication:

- Verifies user identity
- Implemented using login system

Authorization:

- Controls access to resources
- Admin-only routes

Security Techniques:

- Password hashing (bcrypt)
- JWT tokens

4.4.4 Asynchronous Programming

Node.js uses asynchronous programming.

Benefits:

- Handles multiple requests
- Improves performance

4.4.5 Component-Based Architecture (React)

- UI divided into components

- Each component is reusable

Examples:

- Navbar
- Product Card
- Cart

4.4.6 State Management

React manages data using:

- useState
- useEffect

Ensures:

- Dynamic UI updates
- Efficient rendering

4.4.7 Database Design Techniques

- Use of collections instead of tables
- Schema flexibility
- Indexing for performance

4.4.8 Error Handling Techniques

- Try-catch blocks
- Middleware error handling

4.4.9 Testing Techniques

- API testing using Postman
- UI testing using browser
- Debugging using console

4.4.10 Responsive Design

- CSS media queries
- Mobile-friendly design

Advantages of Selected Technologies

Why MERN Stack?

- Full JavaScript stack
- Fast development
- Scalable architecture

Performance Benefits:

- Fast API response
- Efficient frontend rendering

Scalability Benefits:

- Can handle growing users
- Flexible database

CHAPTER 5

DESIGN OF THE PROJECT

5.1 ER/DFD/Flow Chart Diagram

System design defines how different components of the application interact with each other. The Restaurant E-Commerce Website is designed using a modular and layered architecture, ensuring scalability, maintainability, and performance.

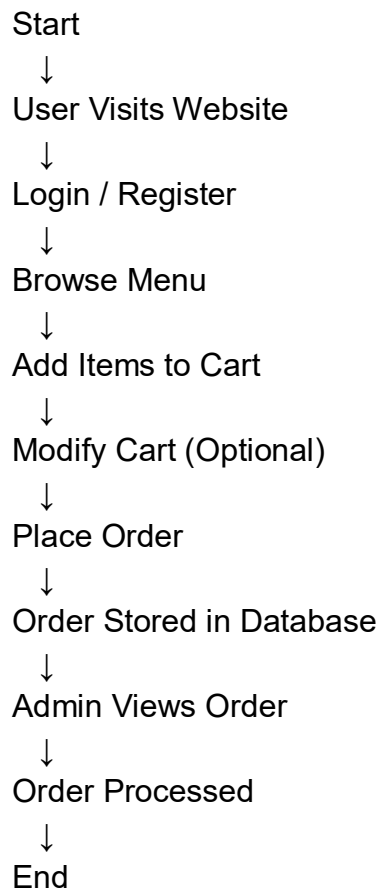
The design focuses on:

- Clear separation of frontend and backend
- Efficient data flow between components
- Easy extensibility for future features

5.1.1 Flow Chart of the System

The flow chart represents the overall working of the system.

Main System Flow



Explanation:

- The system starts when a user accesses the website
- The user must authenticate before placing an order
- The user browses menu items and adds them to cart
- After checkout, the order is stored in the database
- Admin processes the order

5.1.2 Data Flow Diagram (DFD)

DFD shows how data moves through the system.

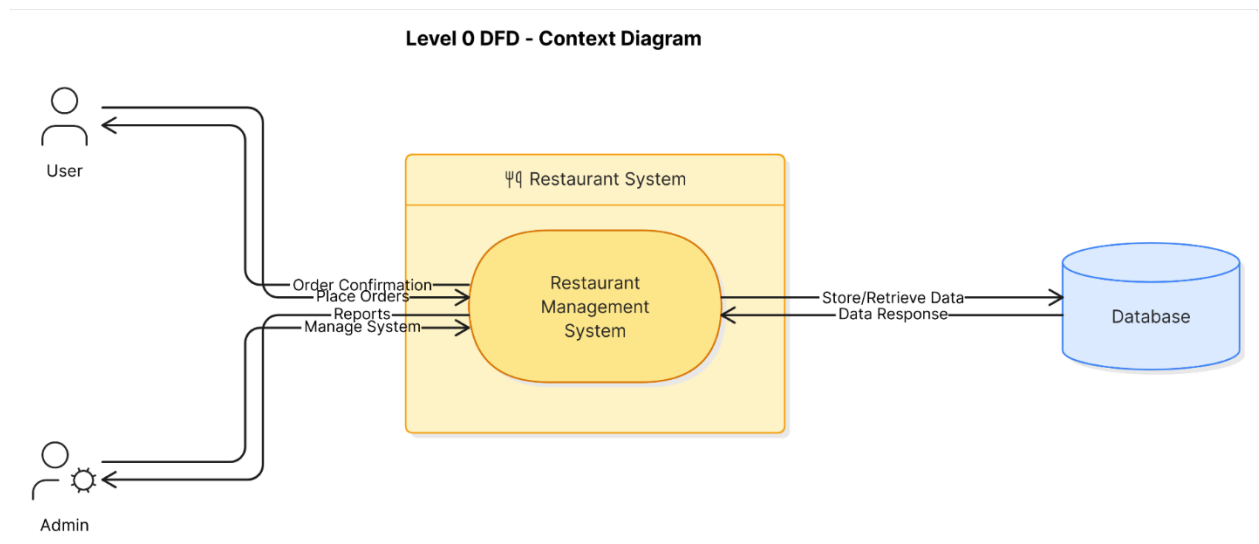
Level 0 DFD (Context Diagram)

[User] -----> (Restaurant System) -----> [Database]

[Admin] -----> (Restaurant System)

Explanation:

- User interacts with system to place orders
- Admin interacts to manage system
- System stores data in database



Level 1 Admin panel DFD

User → Login System → Database

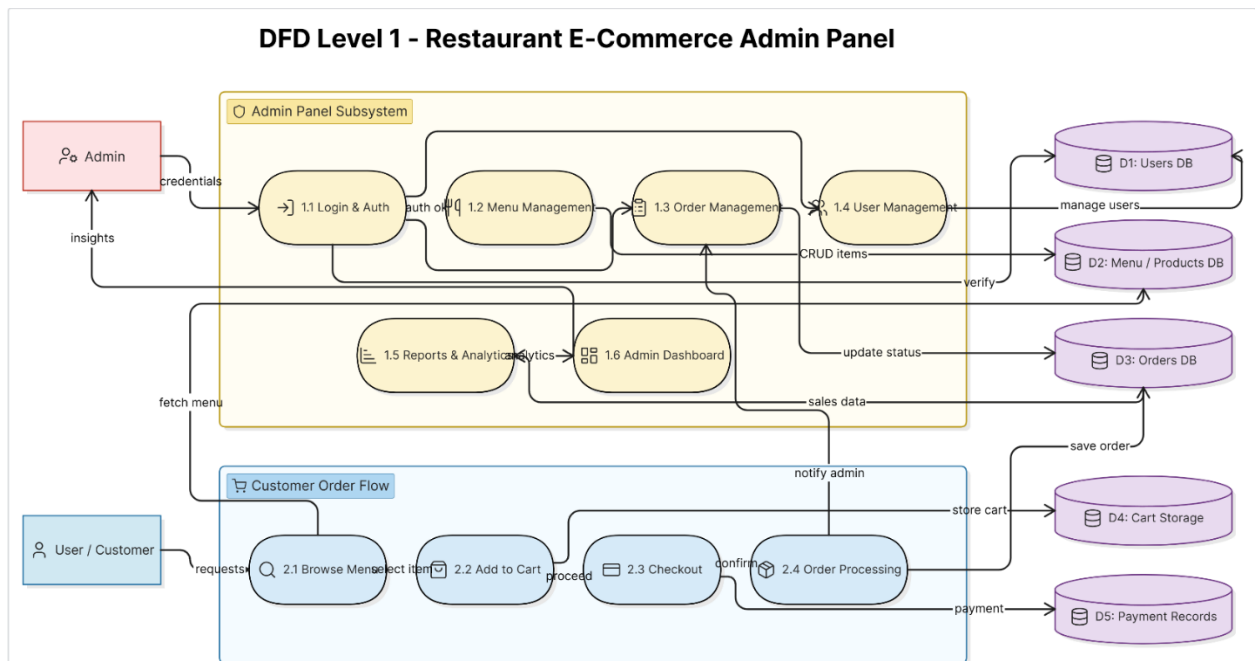
User → Browse Menu → Product DB

User → Cart System → Order System → Database

Admin → Admin Panel → Database

Explanation:

- Login system verifies user
- Menu system fetches products
- Order system stores orders
- Admin panel manages data

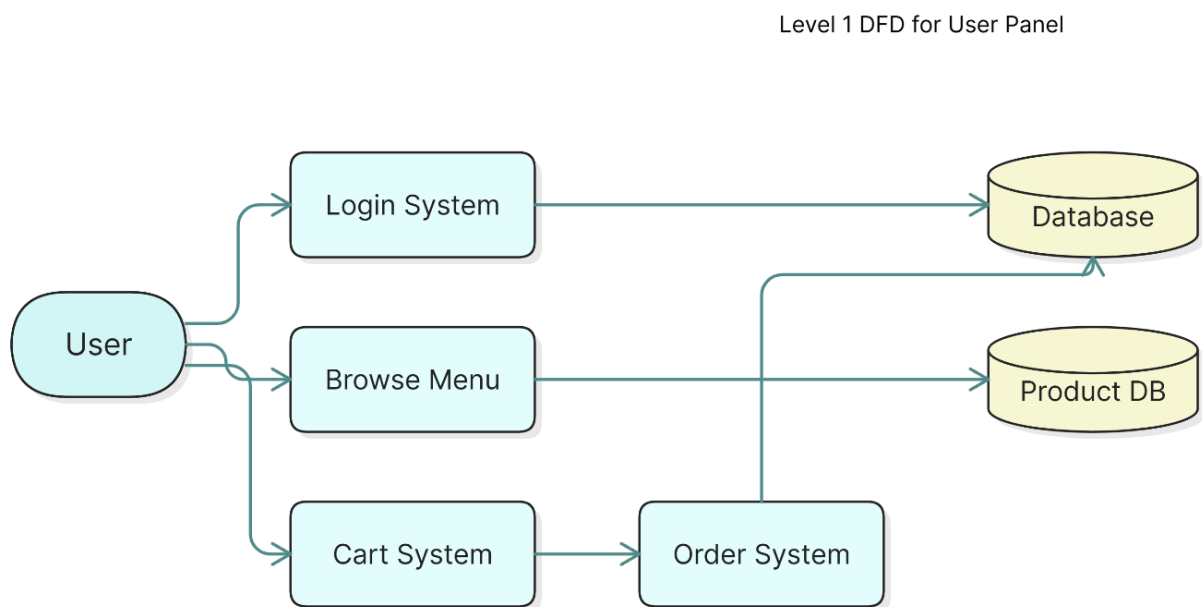


Level 1 User panel DFD (Detailed Order Flow)

User → Add to Cart → Cart Storage
Cart → Checkout → Order Processing
Order Processing → Database
Database → Admin Panel → Admin

Explanation:

- User actions are processed step-by-step
- Data is stored and retrieved efficiently



5.1.3 ER Diagram (Entity Relationship Design)

The ER diagram defines entities and relationships.

Entities:

1. User
2. Product
3. Cart
4. Order

Relationships:

- User → places → Order
- Order → contains → Product
- User → has → Cart

ER Diagram Representation (Text Form)

User (userId, name, email, password)

|

| places

↓

Order (orderId, userId, totalPrice, status)

|

| contains

↓

Product (productId, name, price, category)

User → Cart → Products

Explanation:

- A user can place multiple orders
- Each order contains multiple products
- Each user has a cart

5.2 Class Diagram to show relationship amongst database tables

The class diagram represents object-oriented structure of the system.

Classes and Attributes:

User Class:

- userId
- name
- email
- password

Product Class:

- productId
- name
- category
- price
- description

Order Class:

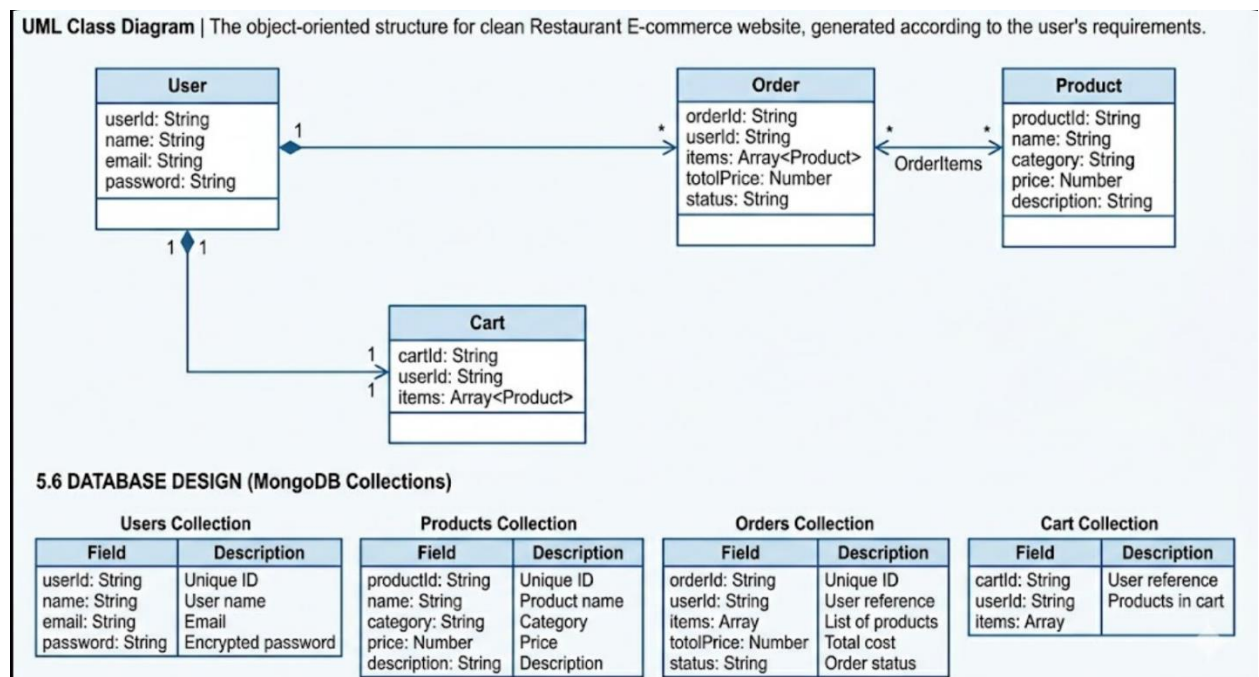
- orderId
- userId
- items
- totalPrice
- status

Cart Class:

- cartId
- userId
- items

Relationships:

- User → Order (One-to-Many)
- Order → Product (Many-to-Many)
- User → Cart (One-to-One)



5.3 Database Tables

The system uses MongoDB collections.

Users Collection

Field	Type	Description
userId	String	Unique ID
name	String	User name
email	String	Email
password	String	Encrypted password

Products Collection

Field	Type	Description
productId	String	Unique ID
name	String	Product name
category	String	Category
price	Number	Price
description	String	Description

Orders Collection

Field	Type	Description
orderId	String	Unique ID
userId	String	User reference
items	Array	List of products
totalPrice	Number	Total cost
status	String	Order status

Cart Collection

Field	Type	Description
userId	String	User reference
items	Array	Products in cart

System Design Advantages

1. Modular Design:

- Easy to maintain
- Easy to extend

2. Scalability:

- Can handle more users
- Can add new features

3. Flexibility:

- MongoDB allows dynamic data

4. Performance:

- Fast API responses
- Efficient data retrieval

Limitations of Design

- No real-time tracking
- No payment integration
- Single restaurant system

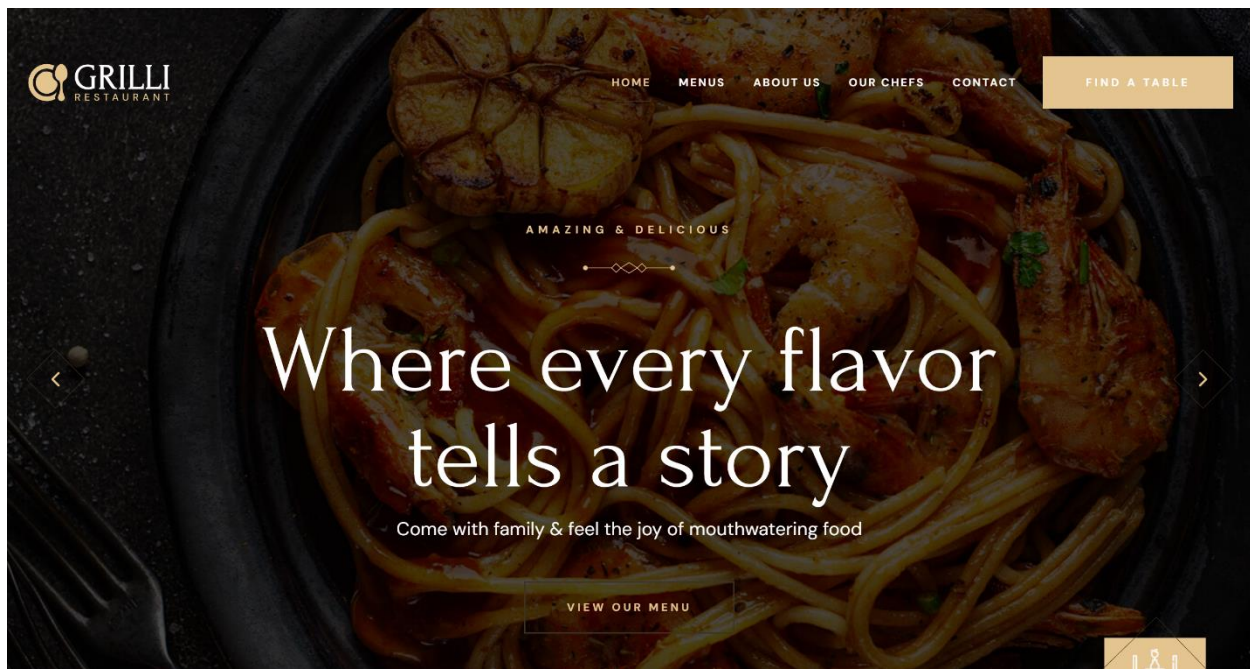
CHAPTER 6

IMPLEMENTATION OF THE PROJECT

6.1 Screenshots of Working Project

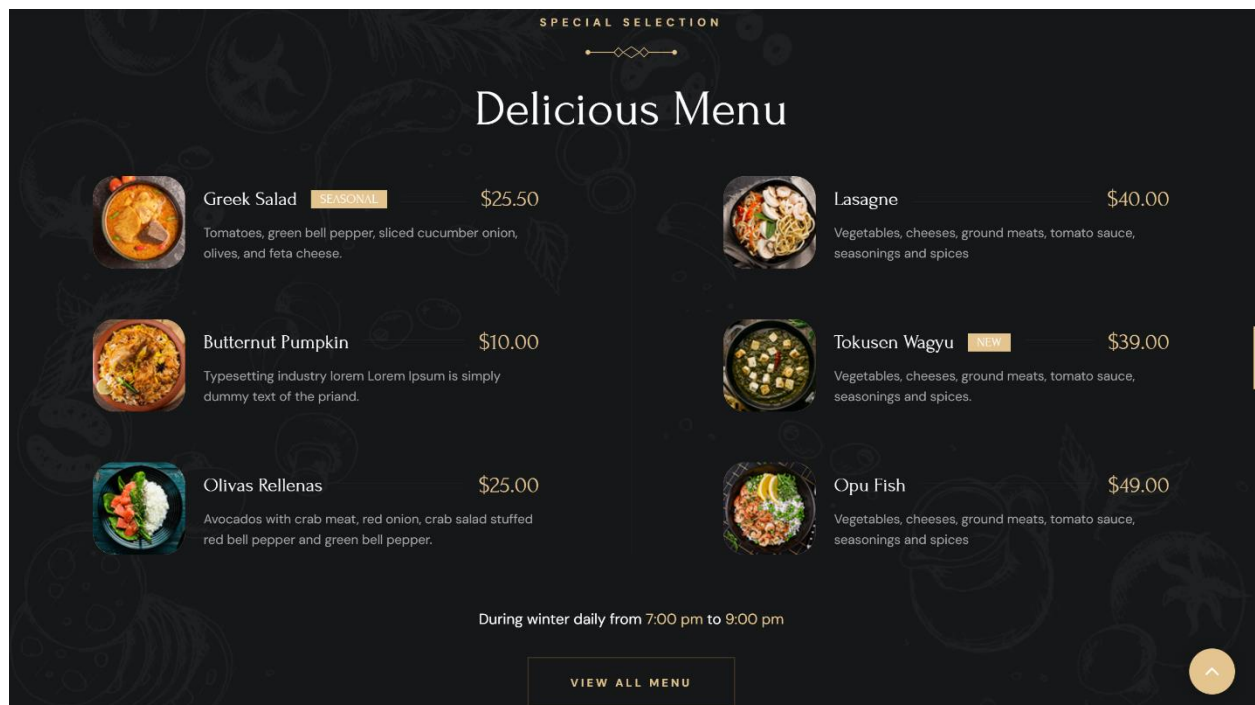
6.1.1 Homepage Interface of the Website

The above screenshot represents the homepage of the Restaurant E-Commerce Website. It serves as the main entry point for users and provides an attractive and user-friendly interface. The homepage features a visually appealing banner with high-quality food imagery to engage users and create a strong first impression. A navigation bar is present at the top, allowing users to easily access different sections such as Home, Menu, About Us, Chefs, and Contact. A prominent call-to-action button like “Find a Table” and “View Our Menu” helps guide users toward key functionalities of the website. The homepage is designed to highlight the theme of the restaurant with a tagline that enhances the overall user experience. It ensures easy navigation, responsiveness, and accessibility, making it convenient for users to explore the website and proceed with ordering food or making reservations.



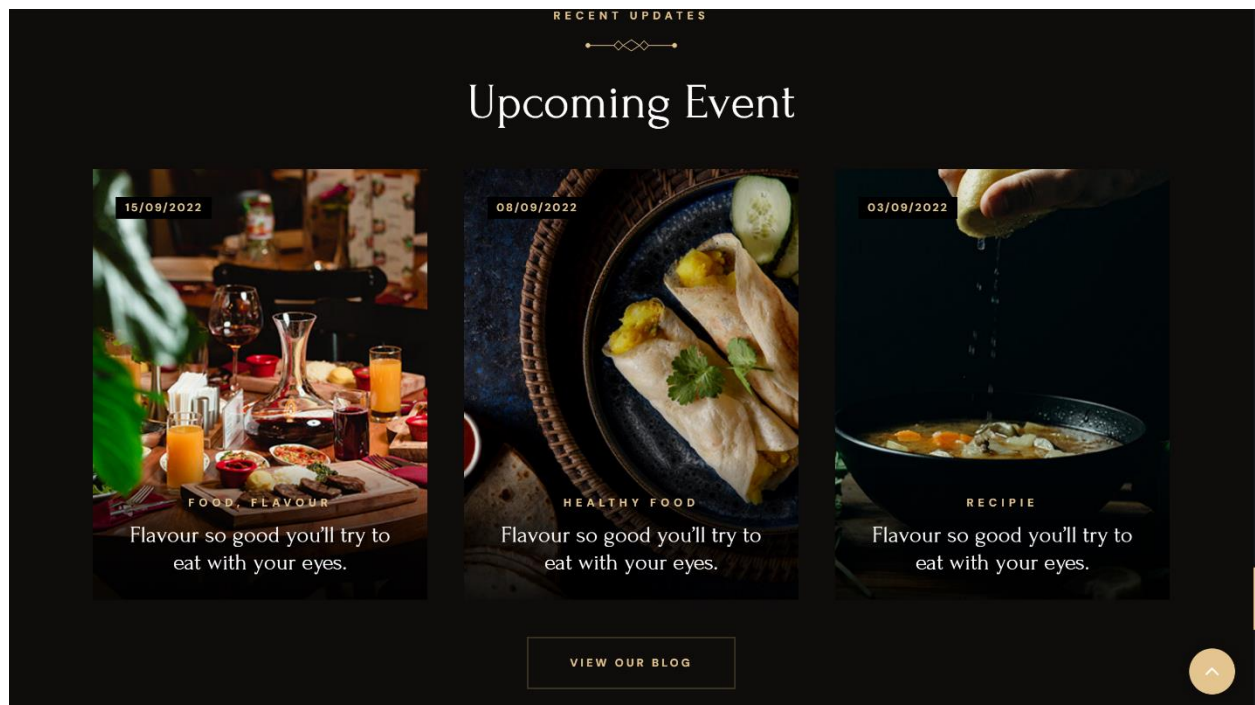
6.1.2 Menu Section of the Website

The above screenshot displays the menu section of the Restaurant E-Commerce Website, where various food items are presented in an organized and visually appealing manner. Each item includes details such as the name of the dish, price, description, and an image to help users make informed choices. The menu is divided into categories and highlights special or new items using labels like “Seasonal” and “New,” enhancing user engagement. The layout is designed to be clean and easy to navigate, allowing users to quickly browse through available dishes. Additionally, a “View All Menu” option is provided to access the complete list of food items. This section plays a crucial role in helping customers explore offerings and select items efficiently for ordering.



6.1.3 Upcoming Events / Blog Section

The above screenshot represents the “Upcoming Events” or blog section of the Restaurant E-Commerce Website. This section is designed to showcase recent updates, events, and featured content related to the restaurant. It displays multiple cards with images, dates, titles, and short descriptions, allowing users to stay informed about new offerings, special events, and food-related updates. The visual layout enhances user engagement by presenting content in an attractive and organized manner. A “View Our Blog” button is also provided to allow users to explore more detailed posts and updates. This section helps in improving user interaction, promoting events, and keeping the website dynamic and informative.



6.1.4 Online Reservation and Contact Section

The above screenshot represents the online reservation and contact section of the Restaurant E-Commerce Website. This section allows users to conveniently book a table by filling out a reservation form with details such as name, phone number, number of persons, date, time, and additional messages. Alongside the reservation form, the contact information of the restaurant is displayed, including phone number, location, and operating hours for lunch and dinner. This provides users with multiple ways to connect with the restaurant. The section is designed to be user-friendly and efficient, enabling quick table bookings and improving customer interaction with the restaurant. It enhances the overall functionality of the website by integrating both reservation and communication features in one place.

The screenshot displays the 'Online Reservation' and 'Contact Us' sections of the Grilli Restaurant website. The 'Online Reservation' section features a form with fields for 'Your Name', 'Phone Number', '1 Person' (dropdown), 'DD-MM-YYYY' (dropdown), '08:00 am' (dropdown), and a 'Message' text area. A 'BOOK A TABLE' button is at the bottom. The 'Contact Us' section displays the 'Booking Request' number '+88-123-123456', the 'Location' 'Restaurant St, Delicious City, London 9578, UK', and operating hours for 'Lunch Time' (Monday to Sunday, 11.00 am - 2.30pm) and 'Dinner Time' (Monday to Sunday, 05.00 pm - 10.00pm). The website has a dark theme with a gold accent color.

GRILLI
RESTAURANT

HOME MENUS ABOUT US OUR CHEFS CONTACT **FIND A TABLE**

Online Reservation

Booking request +88-123-123456 or fill out the order form

Your Name Phone Number

1 Person DD-MM-YYYY 08:00 am

Message

BOOK A TABLE

Contact Us

Booking Request
+88-123-123456

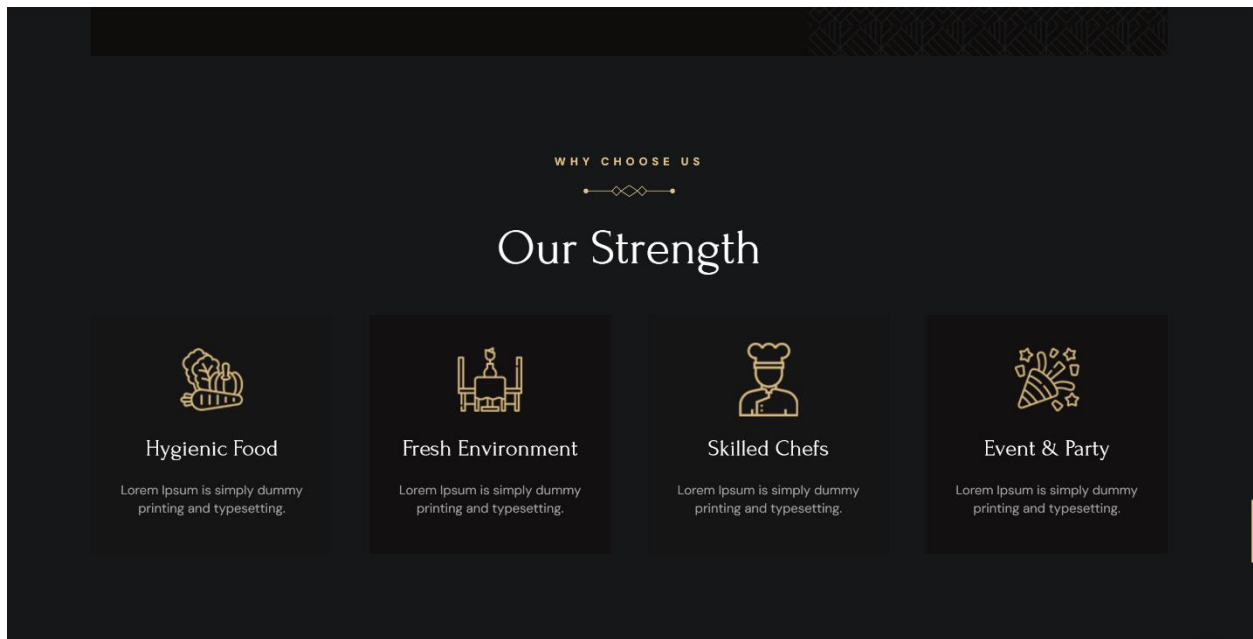
Location
Restaurant St, Delicious City,
London 9578, UK

Lunch Time
Monday to Sunday
11.00 am - 2.30pm

Dinner Time
Monday to Sunday
05.00 pm - 10.00pm

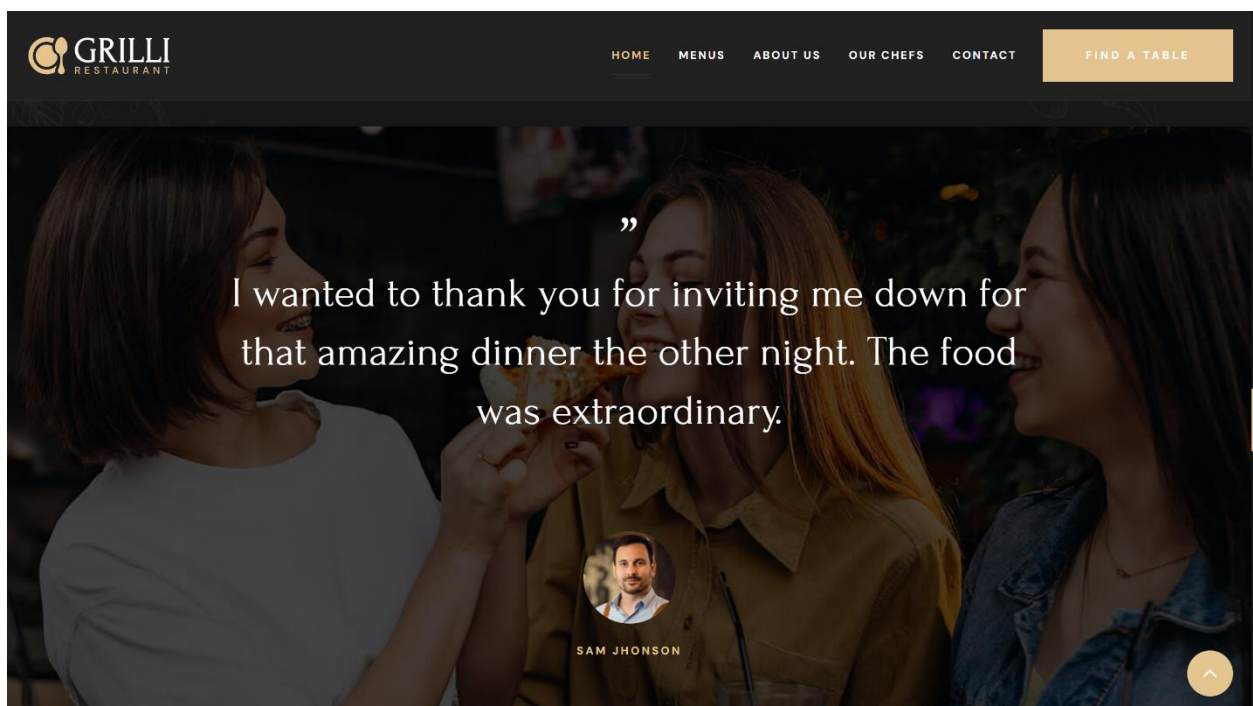
6.1.5 “Why Choose Us” / Our Strength Section

The above screenshot represents the “Why Choose Us” section of the Restaurant E-Commerce Website, which highlights the key strengths and unique features of the restaurant. This section is designed to build trust and attract customers by showcasing the quality of services offered. It includes important aspects such as hygienic food preparation, a fresh and pleasant environment, experienced chefs, and facilities for events and parties. Each feature is represented with icons and short descriptions, making the information easy to understand and visually appealing. This section plays an important role in influencing customer decisions by emphasizing the restaurant’s commitment to quality, service, and customer satisfaction.



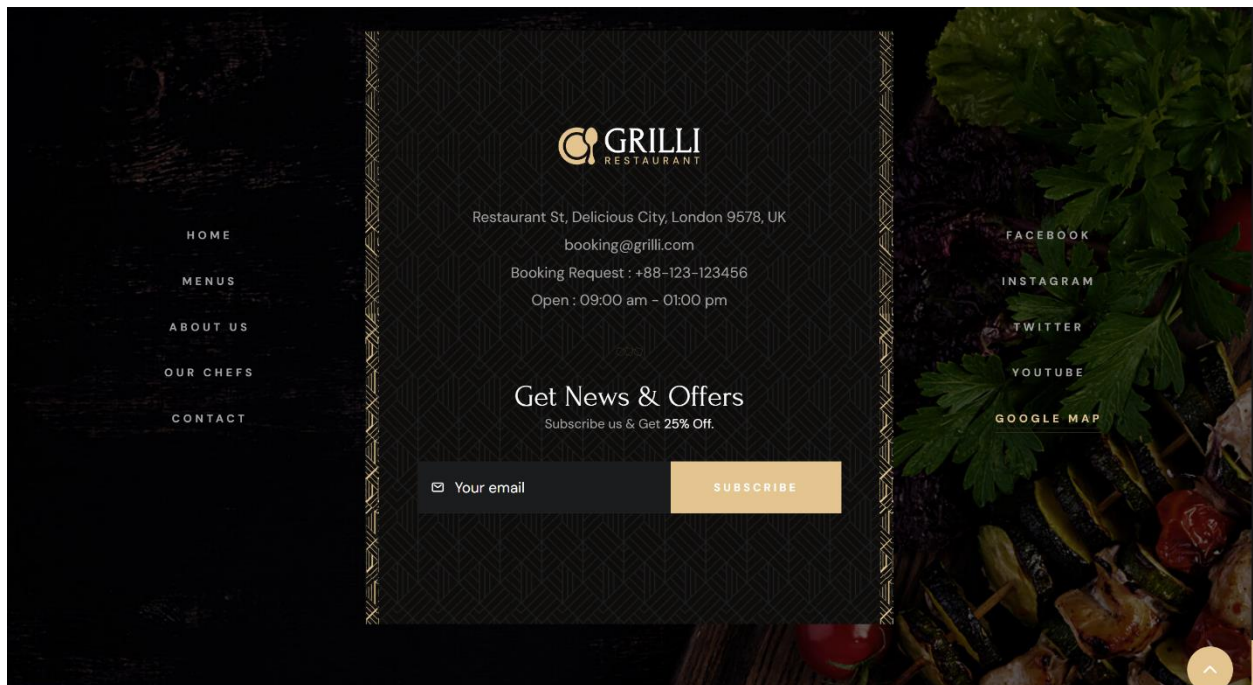
6.1.6 Customer Testimonial Section

The above screenshot represents the customer testimonial section of the Restaurant E-Commerce Website. This section is designed to showcase feedback and reviews from customers to build trust and credibility. It features a highlighted customer quote describing their dining experience, along with the customer's name and profile image. The background image and elegant layout enhance the visual appeal and make the testimonial more engaging. This section plays an important role in influencing potential customers by presenting positive experiences and increasing confidence in the restaurant's quality and service.



6.1.7 Footer Section and Newsletter Subscription

The above screenshot represents the footer section of the Restaurant E-Commerce Website. This section provides essential information about the restaurant, including its address, contact details, email, and operating hours. It also includes a navigation menu for quick access to important pages such as Home, Menu, About Us, Our Chefs, and Contact. Social media links are provided to connect users with the restaurant on various platforms. Additionally, a newsletter subscription feature is available, allowing users to enter their email address to receive updates, news, and special offers. This section enhances user engagement and ensures easy accessibility to important information and services.



6.2 Code Snippets

6.2.1 Parallax Effect JavaScript Implementation

The above screenshot displays a JavaScript code snippet used to implement a parallax scrolling effect on the website. This functionality enhances the visual experience by creating a dynamic movement of elements based on the user's mouse position. The code selects all elements with a specific data attribute (data-parallax-item) and listens for mouse movement events. It then calculates the position of the cursor relative to the screen and adjusts the position of elements accordingly using CSS transformations. By applying different speed values to elements, a layered motion effect is achieved, giving the website a modern and interactive appearance. This improves user engagement and adds a visually appealing animation to the interface.

```
const parallaxItems = document.querySelectorAll("[data-parallax-item]");

let x, y;

window.addEventListener("mousemove", function (event) {

    x = (event.clientX / window.innerWidth * 10) - 5;
    y = (event.clientY / window.innerHeight * 10) - 5;

    // reverse the number eg. 20 -> -20, -5 -> 5
    x = x - (x * 2);
    y = y - (y * 2);

    for (let i = 0, len = parallaxItems.length; i < len; i++) {
        x = x * Number(parallaxItems[i].dataset.parallaxSpeed);
        y = y * Number(parallaxItems[i].dataset.parallaxSpeed);
        parallaxItems[i].style.transform = `translate3d(${x}px, ${y}px, 0px)`;
    }

});
```

6.2.2 Navigation Bar Toggle Functionality

The above screenshot shows a JavaScript implementation used to control the navigation bar toggle functionality of the website. This feature is mainly used for responsive design, especially in mobile view. The code selects elements such as the navigation bar, toggle buttons, and overlay using data attributes. A function is defined to add or remove active classes when the user clicks on the navigation toggle button. This allows the navigation menu to open and close dynamically, improving user experience on smaller screens. The overlay effect also enhances visual clarity by focusing attention on the menu when it is active. Overall, this functionality ensures smooth navigation and better usability across different devices.

```
const navbar = document.querySelector("[data-navbar]");
const navTogglers = document.querySelectorAll("[data-nav-toggler]");
const overlay = document.querySelector("[data-overlay]");

const toggleNavbar = function () {
  navbar.classList.toggle("active");
  overlay.classList.toggle("active");
  document.body.classList.toggle("nav-active");
}

addEventListener(navTogglers, "click", toggleNavbar);
```

6.2.3 Hero Slider Functionality

The above screenshot shows the JavaScript code used to implement a hero slider feature on the website. This functionality allows multiple slides or banners to be displayed dynamically on the homepage. The code selects slider elements and navigation buttons (next and previous) using data attributes. It maintains the current slide position and updates the active

slide by adding or removing CSS classes. Functions such as `slideNext` and `slidePrev` are used to navigate through slides, ensuring a continuous loop from the last slide to the first and vice versa. Event listeners are added to buttons to trigger these transitions on user interaction. This feature enhances the visual appeal of the website and provides an interactive user experience by showcasing important content in a dynamic manner.

```
const heroSlider = document.querySelector("[data-hero-slider]");
const heroSliderItems = document.querySelectorAll("[data-hero-slider-item]");
const heroSliderPrevBtn = document.querySelector("[data-prev-btn]");
const heroSliderNextBtn = document.querySelector("[data-next-btn]");

let currentSlidePos = 0;
let lastActiveSliderItem = heroSliderItems[0];

const updateSliderPos = function () {
  lastActiveSliderItem.classList.remove("active");
  heroSliderItems[currentSlidePos].classList.add("active");
  lastActiveSliderItem = heroSliderItems[currentSlidePos];
}

const slideNext = function () {
  if (currentSlidePos >= heroSliderItems.length - 1) {
    currentSlidePos = 0;
  } else {
    currentSlidePos++;
  }

  updateSliderPos();
}

heroSliderNextBtn.addEventListener("click", slideNext);

const slidePrev = function () {
  if (currentSlidePos <= 0) {
    currentSlidePos = heroSliderItems.length - 1;
  } else {
    currentSlidePos--;
  }
}
```

6.2.4 Online Reservation Form Structure (HTML Implementation)

The above screenshot shows the HTML structure used to design the online reservation form of the Restaurant E-Commerce Website. This form allows users to book a table by entering essential details such as name, phone number, number of persons, date, and time. The form is organized using multiple input fields and dropdown menus, ensuring a clean and user-friendly layout. Icons are integrated alongside input fields to enhance visual clarity and improve the overall user interface. The use of placeholders and structured classes helps in maintaining consistency and responsiveness. Additionally, predefined options for the number of persons and available time slots make the booking process quick and convenient. This implementation plays a key role in enabling efficient table reservation functionality within the website.

```
<section class="reservation">
  <div class="container">

    <div class="form reservation-form bg-black-10">

      <form action="" class="form-left">

        <h2 class="headline-1 text-center">Online Reservation</h2>

        <p class="form-text text-center">
          Booking request <a href="tel:+88123123456" class="link">+88-123-123456</a>
          or fill out the order form
        </p>

        <div class="input-wrapper">
          <input type="text" name="name" placeholder="Your Name" autocomplete="off" class="input-field">

          <input type="tel" name="phone" placeholder="Phone Number" autocomplete="off" class="input-field">
        </div>

        <div class="input-wrapper">

          <div class="icon-wrapper">
            <ion-icon name="person-outline" aria-hidden="true"></ion-icon>

            <select name="person" class="input-field">
              <option value="1-person">1 Person</option>
              <option value="2-person">2 Person</option>
              <option value="3-person">3 Person</option>
              <option value="4-person">4 Person</option>
              <option value="5-person">5 Person</option>
              <option value="6-person">6 Person</option>
              <option value="7-person">7 Person</option>
            </select>


```

```
<ion-icon name="chevron-down" aria-hidden="true"></ion-icon>
</div>

<div class="icon-wrapper">
  <ion-icon name="calendar-clear-outline" aria-hidden="true"></ion-icon>

  <input type="date" name="reservation-date" class="input-field">

  <ion-icon name="chevron-down" aria-hidden="true"></ion-icon>
</div>

<div class="icon-wrapper">
  <ion-icon name="time-outline" aria-hidden="true"></ion-icon>

  <select name="person" class="input-field">
    <option value="08:00am">08 : 00 am</option>
    <option value="09:00am">09 : 00 am</option>
    <option value="010:00am">10 : 00 am</option>
    <option value="011:00am">11 : 00 am</option>
    <option value="012:00am">12 : 00 am</option>
    <option value="01:00pm">01 : 00 pm</option>
    <option value="02:00pm">02 : 00 pm</option>
    <option value="03:00pm">03 : 00 pm</option>
    <option value="04:00pm">04 : 00 pm</option>
    <option value="05:00pm">05 : 00 pm</option>
    <option value="06:00pm">06 : 00 pm</option>
    <option value="07:00pm">07 : 00 pm</option>
    <option value="08:00pm">08 : 00 pm</option>
    <option value="09:00pm">09 : 00 pm</option>
    <option value="10:00pm">10 : 00 pm</option>
  </select>

  <ion-icon name="chevron-down" aria-hidden="true"></ion-icon>
</div>
```

CHAPTER 7

CONCLUSIONS/FUTURE SCOPE OF THE PROJECT

7.1 Conclusions of the Project

The Restaurant E-Commerce Website project was successfully developed to provide a modern, user-friendly, and efficient online platform for restaurant management and food ordering services. The main objective of the project was to simplify the process of browsing menus, placing food orders, managing customer information, and handling restaurant operations digitally. The system helps bridge the gap between customers and restaurants by offering a convenient and time-saving solution.

The project integrates various important functionalities such as user registration and login, menu display, food categorization, cart management, online ordering, and order tracking. An admin panel was also implemented to allow restaurant administrators to manage food items, customer orders, prices, and other essential operations effectively. The website was designed with a responsive interface so that users can access it easily through desktops, tablets, and mobile devices.

During the development of the project, modern web technologies and database management concepts were applied successfully. The project improved practical understanding of frontend development, backend integration, database connectivity, and user interface design. Security and data management techniques were also considered to ensure proper handling of customer information and transactions.

The outcomes of the project demonstrate that the system is capable of reducing manual work, improving order accuracy, and enhancing customer satisfaction. Customers can conveniently explore restaurant menus and place orders from anywhere, while restaurant owners can efficiently manage their services digitally. The project also highlights the growing importance of e-commerce solutions in the food and hospitality industry.

Overall, the Restaurant E-Commerce Website achieved its intended goals and proved to be a reliable, scalable, and effective solution for online food ordering and restaurant management. The project provided valuable technical knowledge and practical experience in software development, making it an important learning achievement in the field of computer science.

7.2 Results in Tabular Form of Project

The results of the Restaurant E-Commerce Website project can be effectively represented using tabular comparison and performance analysis. The tables below highlight the differences between traditional systems and the implemented system, as well as the outcomes achieved.

TABLE 1: Comparison Between Existing System and Proposed System

Parameter	Existing System (Manual/Traditional)	Proposed System (E-Commerce Website)
Order Placement	Manual (phone/in-person)	Online (website-based)
Time Consumption	High	Low
Menu Accessibility	Limited (physical menu)	Available anytime, anywhere
Order Accuracy	Prone to human errors	High accuracy
Payment Method	Mostly cash	Online + Cash options
Record Management	Manual records	Automated database system
Customer Convenience	Limited	High
Scalability	Difficult	Easily scalable

TABLE 2:Functional Results of the System

Module	Expected Outcome	Actual Result Achieved
User Registration	Easy account creation	Successfully implemented
Login System	Secure authentication	Working accurately
Menu Display	Organized food categories	Clear and user-friendly interface
Cart System	Add/remove items easily	Smooth functionality
Order Placement	Quick and efficient ordering	Successfully achieved
Admin Panel	Manage items and orders	Fully functional
Database Handling	Store user and order data	Reliable data storage

TABLE 3:Performance Evaluation

Criteria	Result
System Efficiency	High
User Satisfaction	High
Response Time	Fast
Error Rate	Minimal
Reliability	Strong

Summary of Results

The tabular analysis clearly shows that the proposed Restaurant E-Commerce Website outperforms the traditional system in terms of efficiency, accuracy, and user convenience. The system successfully meets all intended objectives and provides a modern solution for digital food ordering and restaurant management.

7.3 Future Scope of Project

The Restaurant E-Commerce Website developed in this project has significant potential for future enhancements and expansion. Some possible improvements are as follows:

1.Mobile Application Development

In the future, the system can be extended by developing a dedicated mobile application for Android and iOS platforms. This will improve accessibility and provide a more convenient and faster user experience for customers.

2. Online Payment Integration

Advanced and secure online payment options such as UPI, credit/debit cards, and digital wallets can be integrated. This will make transactions more seamless and increase user trust in the platform.

3. Real-Time Order Tracking System

A live order tracking feature can be added, allowing customers to track their food orders in real time. This will enhance transparency and improve customer satisfaction.

4. AI-Based Recommendation System

The system can be upgraded with artificial intelligence to suggest food items based on user preferences, previous orders, and trending items, making the platform more personalized.

5. Multi-Restaurant Support

The website can be expanded into a multi-vendor platform where multiple restaurants can register and offer their services, similar to large food delivery platforms.

These enhancements will make the system more scalable, user-friendly, and competitive in the growing digital food industry. The future scope ensures that the project can evolve into a full-fledged commercial application.

CHAPTER 8

ABOUT THE ORGANISATION



[PG Tech Solutions](#) is a leading IT training and software development company established to provide industry-oriented technical education and advanced digital solutions. They specialize in a wide range of services including Web Development, Mobile Application Development, Digital Marketing, Search Engine Optimization (SEO), Software Development, UI/UX Designing, Cloud Computing, Cyber Security, Artificial Intelligence, Machine Learning, Data Science, Graphic Designing, Video Editing, and IT Consulting Services. PG Tech Solutions also offers Industrial Training, Internship Programs, Corporate Training, Placement Assistance, and Job-Oriented Courses for students and professionals. PG Tech Solutions provides complete IT and software solutions supporting the entire business process from planning and consulting to development, deployment, maintenance, testing, and technical support. With experienced trainers and developers, the company focuses on delivering practical learning, live project exposure, affordable training programs, and career-oriented technical skills to students and businesses. Their goal is to bridge the gap between academic knowledge and real industry requirements through hands-on project-based learning and professional mentorship.

Additionally, PG Tech Solutions provides training programs including 6 Weeks/6 Months Industrial Training, Internship-Based Training, Project-Based Training, Corporate Training, and Job-Oriented Courses Training covering major IT technologies and industry trends such as Full Stack Development (MEAN/MERN), React Js, Angular, Node Js, Vue Js, Python, Artificial Intelligence, Machine Learning, Data Science, Cloud Computing, Cyber Security, Software Testing, Flutter, React Native, Android Development, Java, Advanced Java, C/C++, PHP, Laravel, WordPress, Digital Marketing, SEO, UI/UX Designing, Graphic Designing, Video Editing, AutoCAD, Revit, CATIA, SolidWorks, CNC Programming, English Speaking, Tally, and many more.

The company emphasizes practical implementation, real-time projects, certification programs, interview preparation, and placement support to help students and professionals build successful careers in the IT industry.

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CHAPTER 9

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